

Practice booklet for HKDSE Compulsory Part

Topics included: Percentage

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Designed for Intuitive Building and Rigorous DSE Practice

1 Basic Arithmetic of Percentage

1.1 Percentage as a Unit

Intuition Bridge: Before solving complex word problems, think of the percent symbol “%” as a simple **counting unit**, just like “apples” or “cm”. If you have 5 cm + 3 cm, you simply get 8 cm. Similarly, having 5% simply means **5 units of percent**. Treating percentage as a unit builds immediate confidence without requiring algebraic formulas.

Skill 1.1: Treating Percentage as a Counting Unit

We introduce the percent symbol (%) as a fundamental *unit of measurement*.

- **Definition:** The symbol % represents one hundredth part of a whole quantity.
- **Arithmetic Rule:** When adding or subtracting percentages sharing the same whole, keep the unit % unchanged and perform arithmetic on the numerical coefficients:

$$a\% \pm b\% = (a \pm b)\%$$

- **Key Takeaway:** Treat % as a unit label attached to numbers.

TRAP WARNING: TRAP: DO NOT TREAT THE PERCENT SYMBOL AS AN UNKNOWN ALGEBRAIC VARIABLE LIKE x OR y THAT YOU CAN CANCEL OUT OR MULTIPLY ACCIDENTALLY!

1.2 Addition and Subtraction of Percentage

Intuition Bridge: Adding and subtracting percentages feels completely natural, but there is a golden rule: you can only add or subtract percentages if they belong to the **exact same base quantity** (the same whole “pie”). If you eat 20% of a small pizza and 30% of a huge pizza, you **cannot** simply say you ate 50% of a pizza! Always ensure the reference base is identical before performing arithmetic operations.

Skill 1.2: Addition and Subtraction under a Common Base

We establish the rule for percentage arithmetic under a common base.

- **Common Base Rule:** If $a\%$ and $b\%$ refer to proportions of the *same total quantity*, they can be added or subtracted directly:

$$a\% + b\% = (a + b)\% \quad \text{and} \quad a\% - b\% = (a - b)\%$$

- **Complement Principle:** Since a complete whole is represented by 100%, the remaining percentage after removing parts $a\%$ and $b\%$ is:

$$\text{Remaining Percentage} = 100\% - a\% - b\% = 100\% - (a\% + b\%)$$

TRAP WARNING: TRAP: NEVER ADD OR SUBTRACT PERCENTAGES THAT BELONG TO DIFFERENT BASE QUANTITIES! FOR EXAMPLE, 10% OF CLASS A PLUS 20% OF CLASS B DOES NOT EQUAL 30% OF ALL STUDENTS!

Practical Example 1.2

Problem Statement: A school budget is distributed among three departments. 45% of the total budget is assigned to Mathematics, and 32% is assigned to Science. The rest is assigned to Arts.

- (a) Find the total percentage assigned to Mathematics and Science combined.
- (b) Find the percentage assigned to Arts.

Hint / Key Concept**Common Base Verification**

- Both departments share the *same reference whole* (total school budget).
- The complete pie equals 100%. Subtracting the known parts gives the missing portion.

Solution Steps

- (a) Since both percentages refer to the same total budget:

$$\begin{aligned}\text{Combined \%} &= 45\% + 32\% \\ &= (45 + 32)\% \\ &= \mathbf{77\%}\end{aligned}$$

- (b) The whole budget is 100%.

$$\begin{aligned}\text{Arts \%} &= 100\% - \text{Combined \%} \\ &= 100\% - 77\% \\ &= \mathbf{23\%}\end{aligned}$$

Final Answer: (a) 77%, (b) 23%

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Exercises 1.2**Exercise 1.2.1****Part A (Multiple Choice)**

Evaluate $100\% - (28\% + 37\%)$.

- (A) 35%
- (B) 45%
- (C) 65%
- (D) 73%

Part B (Conventional Question)

Evaluate the expression $85\% - 34\% - 16\%$ step-by-step. Show clearly how terms are grouped.

— Solution Space (Half Page) —

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Exercise 1.2.2**Part A (Multiple Choice)**

If $X = 52\%$ and $Y = 18\%$, find the value of $X - 2Y$.

- (A) 16%
- (B) 26%
- (C) 34%
- (D) 70%

Part B (Conventional Question)

A bakery produces cakes, bread, and cookies daily. 42% of the daily output is bread, and 35% is cookies. The remaining output is cakes.

- (a) Calculate the percentage of daily output that is cakes.
- (b) Find the difference between the percentage of bread and the percentage of cakes produced.

— Solution Space (Half Page) —

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Exercise 1.2.3

Reference: Inspired by HKCEE Mathematics Basic Percentage Distribution Questions

Part A (Multiple Choice)

In a school, 55% of the students are boys. If 22% of the total students in the school are boys who wear glasses, what percentage of the total students are boys who **do not** wear glasses?

- (A) 23%
- (B) 33%
- (C) 45%
- (D) 78%

Part B (Conventional Question)

In an examination, all candidates took both Paper 1 and Paper 2. 15% of the candidates failed Paper 1 only, 10% failed Paper 2 only, and 5% failed both papers.

- (a) Find the percentage of candidates who failed at least one paper.
- (b) Find the percentage of candidates who passed both papers.

— Solution Space (Full Page) —



1.3 Percentage as 1/100

Intuition Bridge: Up to now, we treated % as a unit label like “cm”. But what is 1% mathematically? The word “percent” literally comes from the Latin *per centum*, meaning **out of one hundred**. Thus, the symbol % is simply mathematical shorthand for multiplying by $\frac{1}{100}$ or dividing by 100. Whenever you see %, you can **replace it with $\times \frac{1}{100}$** !

Skill 1.3: Converting Percentage to Fraction and Decimal

We establish the algebraic definition of the percent symbol (%) as a constant numerical factor.

- **Algebraic Definition:** The symbol % is defined as:

$$\% = \frac{1}{100} = 0.01$$

- **Conversion Formula:** To convert $x\%$ to a fraction or decimal, replace % with $\frac{1}{100}$:

$$x\% = x \times \frac{1}{100} = \frac{x}{100}$$

- **Decimal Conversion Rule:** To write $x\%$ as a *decimal number*, divide x by 100 (move the decimal point two places to the left).
- **Key Rule:** Dividing by 100 converts a percentage into a pure numerical value.

TRAP WARNING: TRAP: DO NOT KEEP THE PERCENT SYMBOL AFTER DIVIDING BY 100! WRITE $50\% = \frac{50}{100} = 0.5$, NOT 0.5% . KEEPING THE SYMBOL AFTER CONVERSION CHANGES THE VALUE BY A FACTOR OF 100!

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Practical Example 1.3

Problem Statement: Convert the following percentages into:

- (i) fractions in simplest form, and
(ii) decimals.
(a) 45% (b) 8%

Hint / Key Concept**Conversion Steps**

- Replace % with $\frac{1}{100}$.
- Simplify fractions by dividing top and bottom by their common factor.
- Shift the decimal point **two places to the left** for decimals.

Solution Steps

(a) For 45%:

(i) Fraction = $\frac{45}{100} = \frac{\mathbf{9}}{\mathbf{20}}$

(ii) Decimal = $45 \div 100 = \mathbf{0.45}$

(b) For 8%:

(i) Fraction = $\frac{8}{100} = \frac{\mathbf{2}}{\mathbf{25}}$

(ii) Decimal = $8 \div 100 = \mathbf{0.08}$

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Exercises 1.3**Exercise 1.3.1****Part A (Multiple Choice)**

Express 64% as a fraction in its simplest form.

- (A) $\frac{64}{100}$
- (B) $\frac{32}{50}$
- (C) $\frac{16}{25}$
- (D) $\frac{8}{25}$

Part B (Conventional Question)

Convert each of the following percentages into both a decimal and a fraction in its simplest form:

- (a) 15%
- (b) 84%

— Solution Space (Half Page) —

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Exercise 1.3.2**Part A (Multiple Choice)**

Which of the following numbers has the **largest** value?

- (A) 0.35
- (B) 32%
- (C) $\frac{3}{10}$
- (D) 3.8%

Part B (Conventional Question)

Two students share a pizza. Tom eats 0.25 of the total pizza, while Mary eats 30% of the total pizza.

- (a) Convert 0.25 into a percentage.
- (b) Who ate more pizza? Show your calculations clearly.

— Solution Space (Half Page) —

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Exercise 1.3.3

Reference: Inspired by HKCEE Mathematics Basic Fractional and Percentage Distribution

Part A (Multiple Choice)

A box contains 200 marbles. 35% of the marbles are red, $\frac{2}{5}$ of the marbles are blue, and the remaining marbles are green. What fraction of the marbles in the box are green?

- (A) $\frac{1}{4}$
- (B) $\frac{1}{5}$
- (C) $\frac{3}{10}$
- (D) $\frac{7}{20}$

Part B (Conventional Question)

In a survey of 500 residents, $\frac{3}{10}$ travel to work by bus, 42% travel by MTR, and the remaining residents travel by walking.

- (a) Express $\frac{3}{10}$ as a percentage.
- (b) Find the fraction of residents who travel by walking in its simplest form.
- (c) Calculate the exact number of residents who walk to work.

— Solution Space (Full Page) —

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1.4 Multiplication and Division of Percentage

Intuition Bridge: What does taking 20% **of** 50% mean? In mathematics, the word “of” translates to **multiplication**. Imagine half a pizza (50%). Taking 20% (one-fifth) of that half pizza leaves you with 10% of the whole pizza! Conversely, **division** answers “how many times does one fit into another?”. Dividing 50% by 10% gives 5. Notice how dividing two percentages cancels out the percent units, yielding a **pure scalar number**!

Skill 1.4: Multiplication and Division Involving Percentage

We establish the operational rules for percentage multiplication and percentage division.

- **Multiplication Rule:** To multiply a percentage by a percentage, convert at least one percentage into its *fractional form* or *decimal form* first:

$$a\% \times b\% = \frac{a}{100} \times b\% = \left(\frac{a \times b}{100} \right) \%$$

- **Division Rule:** When dividing a percentage by another percentage, the percent symbols cancel out, leaving a dimensionless ratio:

$$a\% \div b\% = \frac{a\%}{b\%} = \frac{a}{b}$$

- **Key Takeaway:** Multiplication scales a percentage down, while dividing two percentages removes the percent unit completely.

TRAP WARNING: TRAP: DO NOT MULTIPLY THE NUMBERS AND STICK TWO PERCENT SYMBOLS TOGETHER! $20\% \times 50\%$ IS NOT 1000%, AND $20\% \div 10\%$ IS NOT 2%, IT IS THE PURE NUMBER 2!

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Practical Example 1.4**Problem Statement:** Evaluate the following arithmetic expressions:

(a) $20\% \times 30\%$

(b) $45\% \div 15\%$

Hint / Key Concept**Operation Rules**

- **Multiplication:** Convert one % symbol to $\frac{1}{100}$ before multiplying.
- **Division:** Percent units cancel out: $\frac{\%}{\%} = 1$. The result has no % unit!

Solution Steps(a) For $20\% \times 30\%$:Convert 20% to fraction: $\frac{20}{100}$

$$\begin{aligned}\text{Expression} &= \frac{20}{100} \times 30\% \\ &= \frac{1}{5} \times 30\% \\ &= 6\%\end{aligned}$$

(b) For $45\% \div 15\%$:

$$\text{Expression} = \frac{45\%}{15\%}$$

$$\text{Cancel the \% symbol: } = \frac{45}{15} = 3$$

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Exercises 1.4**Exercise 1.4.1****Part A (Multiple Choice)**

Evaluate $40\% \times 50\%$.

- (A) 2%
- (B) 20%
- (C) 200%
- (D) 2000%

Part B (Conventional Question)

Evaluate the following expressions step-by-step:

- (a) $25\% \times 40\%$
- (b) $80\% \div 20\%$

— Solution Space (Half Page) —

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Exercise 1.4.2**Part A (Multiple Choice)**

What is 30% of 60%?

- (A) 1.8%
- (B) 18%
- (C) 90%
- (D) 180%

Part B (Conventional Question)

In a school, 60% of the students are girls. Among the girls, 50% wear glasses.

- (a) Express 50% as a fraction in its simplest form.
- (b) Find the percentage of the total students in the school who are girls wearing glasses.

— Solution Space (Half Page) —

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Exercise 1.4.3

Reference: Inspired by HKCEE Mathematics Paper 2 Multi-stage Percentage Problems

Part A (Multiple Choice)

A container is 80% full of liquid. If 25% of the liquid in the container is poured out, what percentage of the container's total capacity was poured out?

- (A) 20%
- (B) 25%
- (C) 55%
- (D) 60%

Part B (Conventional Question)

A student spent 10% of his monthly savings on a shirt and 20% of his monthly savings on trousers. He then spent 30% of his *remaining savings* on books.

- (a) Find the percentage of his monthly savings remaining after buying the shirt and trousers.
- (b) Find the percentage of his total monthly savings spent on books.
- (c) Find the total percentage of his monthly savings spent altogether.

— Solution Space (Full Page) —

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1.5 Arithmetic on Expressions Involving Percentage

Intuition Bridge: What happens when percentage symbols are mixed with ordinary integers, fractions, or algebraic operations in a single expression? The fundamental rule of mathematics never changes: **order of operations (BODMAS/PEMDAS)** still rules! The easiest way to handle mixed expressions is to **convert all percentages into fractions or decimals** before performing arithmetic with regular numbers.

Skill 1.5: Evaluating Mixed Percentage Expressions

We establish the operational rules for evaluating a mixed percentage expression containing numbers, fractions, and percent symbols.

- **Definition:** A mixed percentage expression combines standard arithmetic operations ($+$, $-$, $\times$, $\div$) with both percentage values and pure numerical values.
- **Evaluation Strategy:**
 1. Perform operations inside **brackets** first.
 2. Convert percentages to *fractions* ($\frac{x}{100}$) or *decimals* ($0.01x$) when operating alongside pure numbers.
 3. Execute **multiplication and division** from left to right, followed by **addition and subtraction**.
- **Key Rule:** Never add a pure number directly to a percentage without converting first.

TRAP WARNING: TRAP: DO NOT ADD A PURE NUMBER DIRECTLY TO A PERCENTAGE! FOR EXAMPLE, $100 + 20\%$ IS NOT 120% , NOR IS IT 120 . IT EQUALS $100 + 0.2 = 100.2$!

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Practical Example 1.5

Problem Statement: Evaluate the following mixed arithmetic expression:

$$120 \times 25\% - (40\% \div 8\%)$$

Hint / Key Concept**Order of Operations**

- **Brackets:** $40\% \div 8\% = 5$ (percent units cancel out).
- **Multiplication:** $120 \times \frac{1}{4} = 30$.
- **Subtraction:** $30 - 5 = 25$.

Solution Steps

1. Evaluate the division inside brackets:

$$(40\% \div 8\%) = \frac{40}{8} = 5$$

2. Convert 25% to a fraction and multiply:

$$120 \times 25\% = 120 \times \frac{25}{100} = 120 \times \frac{1}{4} = 30$$

3. Subtract the results:

$$\text{Expression} = 30 - 5 = \mathbf{25}$$

Final Answer: 25

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Exercises 1.5**Exercise 1.5.1****Part A (Multiple Choice)**

Evaluate $15 + 40\% \times 50$.

- (A) 20
- (B) 35
- (C) 275
- (D) 2750

Part B (Conventional Question)

Evaluate the expression $200 \times 15\% - (50\% \div 10\%)$ step-by-step. Show all intermediate steps clearly.

— Solution Space (Half Page) —

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Exercise 1.5.2**Part A (Multiple Choice)**

If $K = 150 \times 40\% - 80 \times 25\%$, find the value of K .

- (A) 20
- (B) 40
- (C) 60
- (D) 80

Part B (Conventional Question)

A crate contains 180 oranges. 25% of the oranges are sold in the morning, and 15 oranges are sold in the afternoon.

- (a) Find the number of oranges sold in the morning.
- (b) Find the total number of oranges remaining in the crate.

— Solution Space (Half Page) —

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Exercise 1.5.3

Reference: Inspired by HKCEE Mathematics Paper 2 Basic Mixed Arithmetic Review

Part A (Multiple Choice)

Express the value of $\frac{1}{2} + 25\% \times 1.6$ as a percentage.

- (A) 40%
- (B) 65%
- (C) 90%
- (D) 140%

Part B (Conventional Question)

A student has a monthly allowance of \$600. He spends 30% of his allowance on transportation and 25% on food. From the remaining money, he donates \$50 to a charity.

- (a) Calculate the total amount spent on transportation and food.
- (b) Find the amount of money he has left after the donation.
- (c) Express the final remaining amount of money as a percentage of his original monthly allowance.

— Solution Space (Full Page) —

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1.6 Powers of Percentages

Intuition Bridge: What happens when you square a percentage, like $(20\%)^2$? A common mistake is to square only the number and keep the percentage sign, giving 400%. But remember: **the percentage sign % is a unit meaning $\frac{1}{100}$!** When you square a percentage, **you must square both the number AND the unit $\frac{1}{100}$.** Always convert the percentage to a fraction or decimal first before applying exponents!

Skill 1.6: Evaluating Powers of Percentages

We establish the rule for taking powers of percentage values.

- **General Rule:** To calculate $(r\%)^n$:

Step 1: Convert to decimal/fraction $\rightarrow r\% = \frac{r}{100}$

Step 2: Apply the power $\rightarrow \left(\frac{r}{100}\right)^n$

Step 3: Convert back to percentage $\rightarrow \left(\frac{r}{100}\right)^n \times 100\%$

- **Key Conversion Examples:**

- $(10\%)^2 = (0.1)^2 = 0.01 = \mathbf{1\%}$
- $(20\%)^2 = (0.2)^2 = 0.04 = \mathbf{4\%}$
- $(50\%)^2 = (0.5)^2 = 0.25 = \mathbf{25\%}$

TRAP WARNING: TRAP: $(x\%)^n \neq x^2\%$! FOR EXAMPLE, $(30\%)^2 \neq 900\%$. INSTEAD, $(0.3)^2 = 0.09 = 9\%$! NEVER KEEP THE % SIGN UNCHANGED WHEN TAKING POWERS!

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Practical Example 1.6

Problem Statement: Evaluate the following expressions and express each answer as a percentage:

(a) $(30\%)^2$

(b) $(20\%)^3 + (40\%)^2$

Hint / Key Concept**Fraction View**

- $(30\%)^2 = \left(\frac{3}{10}\right)^2 = \frac{9}{100} = 9\%$.
- $(20\%)^3 = \left(\frac{1}{5}\right)^3 = \frac{1}{125} = 0.008 = 0.8\%$.
- Working in decimals or fractions avoids losing powers of 10!

Solution Steps

(a) Calculate $(30\%)^2$:

$$30\% = 0.3$$

$$(30\%)^2 = (0.3)^2 = 0.09$$

As a percentage $= 0.09 \times 100\% = \mathbf{9\%}$

(b) Calculate $(20\%)^3 + (40\%)^2$:

$$20\% = 0.2, \quad 40\% = 0.4$$

$$(20\%)^3 = (0.2)^3 = 0.008$$

$$(40\%)^2 = (0.4)^2 = 0.16$$

$$\text{Sum} = 0.008 + 0.16 = 0.168$$

As a percentage $= 0.168 \times 100\% = \mathbf{16.8\%}$

Final Answer: (a) 9%, (b) 16.8%

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Exercises 1.6**Exercise 1.6.1****Part A (Multiple Choice)**

Which of the following is equal to $(40\%)^2$?

- (A) 0.16%
- (B) 1.6%
- (C) 16%
- (D) 1600%

Part B (Conventional Question)

Evaluate each of the following expressions and express your answers as percentages:

- (a) $(10\%)^3$
- (b) $(60\%)^2 - (30\%)^2$
- (c) $\frac{(50\%)^2}{10\%}$

— Solution Space (Half Page) —

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Exercise 1.6.2**Part A (Multiple Choice)**

Find the value of $(80\%)^2 + (60\%)^2$.

- (A) 10%
- (B) 100%
- (C) 280%
- (D) 10000%

Part B (Conventional Question)

The length of each side of a square is reduced to 70% of its original length.

- (a) Express the new side length as a fraction of the original side length.
- (b) Express the new area of the square as a percentage of the original area using powers of percentages.
- (c) Calculate the percentage decrease in the area of the square.

— Solution Space (Half Page) —

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Exercise 1.6.3

Reference: Inspired by HKDSE Foundation Concepts / Numerical Reasoning

Part A (Multiple Choice)

If $x = 20\%$, find the value of $\frac{x^2+x}{x}$.

- (A) 24%
- (B) 104%
- (C) 120%
- (D) 600%

Part B (Conventional Question)

Consider the expression $E = \frac{(30\%)^2 \times (40\%)}{(20\%)^3}$.

- (a) Convert each component percentage into a decimal.
- (b) Simplify and calculate the exact value of E .
- (c) Express E as a percentage.

— Solution Space (Full Page) —

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1.7 Further Arithmetic on Percentages

Intuition Bridge: Now that we can perform addition, multiplication, division, and exponents on percentages, we combine them into multi-step algebraic expressions! The most dangerous operation in percentage arithmetic is taking **square roots**. For example, what is $\sqrt{16\%}$? It is NOT 4%! **Since $16\% = 0.16$, taking the square root gives $\sqrt{0.16} = 0.4 = 40\%$.** Always convert to standard decimals or fractions first!

Skill 1.7: Order of Operations and Roots with Percentages

We establish the protocol for complex expressions involving square roots and mixed operations with percentages.

- **Square Root Rule for Percentages:**

$$\sqrt{r\%} = \sqrt{\frac{r}{100}} = \frac{\sqrt{r}}{10} = \left(\frac{\sqrt{r}}{10} \times 100\right)\%$$

- $\sqrt{16\%} = \sqrt{0.16} = 0.4 = \mathbf{40\%}$
- $\sqrt{25\%} = \sqrt{0.25} = 0.5 = \mathbf{50\%}$
- $\sqrt{81\%} = \sqrt{0.81} = 0.9 = \mathbf{90\%}$

- **Order of Operations (PEMDAS/BODMAS):**

1. Convert all percentages into decimals or fractions ($\frac{1}{100}$).
2. Evaluate Roots and Powers first.
3. Perform Multiplication and Division from left to right.
4. Perform Addition and Subtraction from left to right.
5. Convert the final result to the required form (decimal, fraction, or percentage).

TRAP WARNING: TRAP: $\sqrt{x\%} \neq \sqrt{x}\%$! $\sqrt{16\%} = 40\%$, NOT 4%! TAKING A SQUARE ROOT OF A NUMBER SMALLER THAN 1 MAKES IT BIGGER, NOT SMALLER!

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Practical Example 1.7

Problem Statement: Evaluate the expression and express the result as a percentage:

$$\frac{\sqrt{64\%} - (40\%)^2}{20\%}$$

Hint / Key Concept**Sanity Check**

- $\sqrt{64\%} = 80\%$.
- $(40\%)^2 = 16\%$.
- $80\% - 16\% = 64\%$.
- $\frac{64\%}{20\%} = \frac{0.64}{0.20} = 3.2 = 320\%$.
- Note that dividing a percentage by a percentage yields a pure number!

Solution Steps

1. Convert components to decimals:

$$\sqrt{64\%} = \sqrt{0.64} = 0.8$$

$$(40\%)^2 = (0.4)^2 = 0.16$$

$$20\% = 0.2$$

2. Evaluate the numerator:

$$\text{Numerator} = 0.8 - 0.16 = 0.64$$

3. Divide by denominator:

$$\text{Value} = \frac{0.64}{0.2} = 3.2$$

4. Convert to percentage:

$$3.2 \times 100\% = \mathbf{320\%}$$

Final Answer: 320%

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Exercises 1.7**Exercise 1.7.1****Part A (Multiple Choice)**

Find the value of $\sqrt{49\%} + 30\%$.

- (A) 37%
- (B) 40%
- (C) 100%
- (D) 370%

Part B (Conventional Question)

Evaluate the following expressions and present each final answer as a percentage:

- (a) $\sqrt{36\%} \times 50\%$
- (b) $\sqrt{100\% - 19\%}$
- (c) $\frac{\sqrt{9\%} + (10\%)^2}{4\%}$

— Solution Space (Half Page) —

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Exercise 1.7.2**Part A (Multiple Choice)**

Which of the following expressions gives the LARGEST value?

- (A) $(50\%)^2$
- (B) $50\% \times 50\%$
- (C) $\sqrt{50\%}$
- (D) $50\% \div 2$

Part B (Conventional Question)

Given that $A = \sqrt{25\%}$ and $B = (30\%)^2$.

- (a) Express A as a percentage.
- (b) Express B as a percentage.
- (c) Find the value of $\frac{A-B}{A+B}$, correct to 2 decimal places.

— Solution Space (Half Page) —

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Exercise 1.7.3

Reference: Inspired by HKDSE Paper 2 Quantitative Reasoning Questions

Part A (Multiple Choice)

If $\sqrt{x\%} = 60\%$, find the value of x .

- (A) 0.36
- (B) 3.6
- (C) 36
- (D) 360

Part B (Conventional Question)

Solve for the positive unknown k in each of the following percentage equations:

- (a) $\sqrt{k\%} + 20\% = 70\%$
- (b) $\frac{(k\%)^2}{\sqrt{16\%}} = 9\%$
- (c) Express $(k\%)^3$ as a percentage using the value of k obtained from part (b).

— Solution Space (Full Page) —

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2 Representation Using Percentage

2.1 Increase and Percentage Increase

Intuition Bridge: Imagine you have \$100 pocket money and your parents give you \$20 extra. The increase is the actual dollar amount (\$20), while the percentage increase compares that extra amount to what you started with (20%). Always ask yourself: "Compared to what?" The starting point (**original value**) is always your 100% base reference!

Skill 2.1: Finding Increase and Percentage Increase

We establish the mathematical definitions for increase and percentage increase.

- **Increase:** The actual amount added to an original quantity:

$$\text{Increase} = \text{New Value} - \text{Original Value}$$

- **Percentage Increase:** The ratio of the increase relative to the original value, expressed as a percentage:

$$\text{Percentage Increase} = \frac{\text{Increase}}{\text{Original Value}} \times 100\% = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100\%$$

- **New Value Formula:** To find the new value directly after an increase of $r\%$:

$$\text{New Value} = \text{Original Value} \times (1 + r\%)$$

TRAP WARNING: TRAP: ALWAYS DIVIDE BY THE ORIGINAL VALUE WHEN CALCULATING PERCENTAGE INCREASE! NEVER DIVIDE BY THE NEW VALUE!

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Practical Example 2.1

Problem Statement: The price of a school bag was \$80. Due to inflation, its price increased to \$100.

- (a) Find the increase in price.
- (b) Find the percentage increase in price.

Hint / Key Concept**Base Comparison**

- **Original Value:** \$80 is the base (100%).
- **Denominator:** Divide by \$80, NOT \$100.
- **Check:** \$20 is one-quarter of \$80, which is 25%.

Solution Steps

- (a) Calculate the actual increase:

$$\begin{aligned}\text{Increase} &= \text{New Price} - \text{Original Price} \\ &= \$100 - \$80 = \mathbf{\$20}\end{aligned}$$

- (b) Calculate the percentage increase:

$$\begin{aligned}\text{Percentage Increase} &= \frac{\text{Increase}}{\text{Original Price}} \times 100\% \\ &= \frac{20}{80} \times 100\% \\ &= \frac{1}{4} \times 100\% = \mathbf{25\%}\end{aligned}$$

Final Answer: (a) \$20, (b) 25%

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercises 2.1**Exercise 2.1.1****Part A (Multiple Choice)**

The length of a plant grew from 20 cm to 25 cm. Find the percentage increase in length.

- (A) 20%
- (B) 25%
- (C) 80%
- (D) 125%

Part B (Conventional Question)

The weight of a parcel increased from 4 kg to 5 kg.

- (a) Find the increase in weight.
- (b) Find the percentage increase in weight.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.1.2**Part A (Multiple Choice)**

A worker's daily wage was \$500. After a wage review, his wage was increased by 12%. Find his new daily wage.

- (A) \$512
- (B) \$560
- (C) \$600
- (D) \$620

Part B (Conventional Question)

The monthly allowance of a student was \$400. His allowance is increased by 15% this month.

- (a) Calculate the increase in his monthly allowance.
- (b) Find his new monthly allowance using two different methods:
 - (i) Adding the increase to the original allowance.
 - (ii) Multiplying directly by $(1 + 15\%)$.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.1.3

Reference: Inspired by HKCEE 1989 MA 1-1 / HKCEE 2002 MA 1-6[cite: 2]

Part A (Multiple Choice)

The monthly income of a worker is increased from \$8000 to \$9000[cite: 2]. Find the percentage increase in his monthly income.

- (A) 11.1%
- (B) 12.5%
- (C) 88.9%
- (D) 112.5%

Part B (Conventional Question)

The radius of a circular disk is 8 cm[cite: 2]. A new circular disk is formed by increasing the radius by 10%[cite: 2].

- (a) Find the radius of the new circular disk.
- (b) Express the area of the original disk and the new disk in terms of π .
- (c) Find the percentage increase in the area of the circular disk[cite: 2].

— Solution Space (Full Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

2.2 Decrease and Percentage Decrease

Intuition Bridge: Just as prices or quantities can grow, they can also shrink. If a item originally priced at \$100 is now sold for \$80, the decrease is the actual reduction amount (\$20), while the percentage decrease compares that reduction back to where it started (20%). Notice that whether a quantity increases or decreases, the **original starting value** is **always the baseline (100% denominator)**!

Skill 2.2: Finding Decrease and Percentage Decrease

We establish the mathematical definitions for decrease and percentage decrease.

- **Decrease:** The actual amount reduced from an original quantity:

$$\text{Decrease} = \text{Original Value} - \text{New Value}$$

- **Percentage Decrease:** The ratio of the decrease relative to the original value, expressed as a percentage:

$$\text{Percentage Decrease} = \frac{\text{Decrease}}{\text{Original Value}} \times 100\% = \frac{\text{Original Value} - \text{New Value}}{\text{Original Value}} \times 100\%$$

- **New Value Formula:** To find the new value directly after a decrease of $r\%$:

$$\text{New Value} = \text{Original Value} \times (1 - r\%)$$

TRAP WARNING: TRAP: ALWAYS USE THE ORIGINAL STARTING VALUE AS THE DENOMINATOR! NEVER DIVIDE BY THE NEW SMALLER VALUE WHEN CALCULATING PERCENTAGE DECREASE!

MATHS

CLARITY | CONFIDENCE | SUCCESS

Practical Example 2.2

Problem Statement: A jacket originally costs \$200. During a clearance sale, its price is reduced to \$150.

- (a) Find the decrease in price.
 (b) Find the percentage decrease in price.

Hint / Key Concept**Base Comparison**

- **Original Value:** \$200 is the starting baseline (100%).
- **Denominator:** Divide by \$200, NOT \$150.
- **Check:** \$50 reduction out of \$200 equals 25%.

Solution Steps

- (a) Calculate the actual decrease:

$$\begin{aligned}\text{Decrease} &= \text{Original Price} - \text{New Price} \\ &= \$200 - \$150 = \mathbf{\$50}\end{aligned}$$

- (b) Calculate the percentage decrease:

$$\begin{aligned}\text{Percentage Decrease} &= \frac{\text{Decrease}}{\text{Original Price}} \times 100\% \\ &= \frac{50}{200} \times 100\% \\ &= \frac{1}{4} \times 100\% = \mathbf{25\%}\end{aligned}$$

Final Answer: (a) \$50, (b) 25%

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercises 2.2**Exercise 2.2.1****Part A (Multiple Choice)**

The weight of a bag of flour decreased from 10 kg to 8 kg. Find the percentage decrease in weight.

- (A) 2%
- (B) 20%
- (C) 25%
- (D) 80%

Part B (Conventional Question)

The price of a book dropped from \$50 to \$40.

- (a) Find the decrease in price.
- (b) Find the percentage decrease in price.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.2.2**Part A (Multiple Choice)**

An item costing \$400 is on sale at a price reduction of 15%. Find its new price.

- (A) \$60
- (B) \$340
- (C) \$360
- (D) \$385

Part B (Conventional Question)

A school club had 250 members last year. The number of members decreased by 8% this year.

- (a) Calculate the decrease in the number of members.
- (b) Find the new number of members in the school club using two different methods:
 - (i) Subtracting the decrease from the original number of members.
 - (ii) Multiplying directly by $(1 - 8\%)$.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.2.3

Reference: Inspired by HKCEE 2009 MA 1-7 / HKCEE 2010 MA 1-7[cite: 2]

Part A (Multiple Choice)

Mary has 50 badges. The number of badges owned by Tom is 30% less than that owned by Mary[cite: 2]. How many badges does Tom have?

- (A) 15
- (B) 20
- (C) 35
- (D) 65

Part B (Conventional Question)

In a survey, there are 172 male interviewees[cite: 2]. The number of female interviewees is 75% less than that of male interviewees[cite: 2].

- (a) Find the number of female interviewees[cite: 2].
- (b) Find the total number of interviewees in the survey.
- (c) Calculate the percentage of female interviewees in the survey[cite: 2].

— Solution Space (Full Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

2.3 Increase and Decrease as Change

Intuition Bridge: Instead of keeping track of “increase” and “decrease” as two completely separate ideas with different subtraction orders, mathematicians combine them into one unified concept called *change*. Think of change like moving along a number line: moving forward means a positive change (increase), while moving backward means a negative change (decrease).

Skill 2.3: Representing Change as a Signed Quantity

We establish the unified mathematical definition for change.

- **Definition of Change:** The change in a quantity is always calculated by subtracting the original value from the new value:

$$\text{Change} = \text{New Value} - \text{Original Value}$$

- **Direction of Change:**
 - If $\text{Change} > 0$ (positive), it represents an increase.
 - If $\text{Change} < 0$ (negative), it represents a decrease.
- **Sign Convention:** A change of $-\$20$ is mathematically identical to a decrease of $\$20$.

TRAP WARNING: TRAP: DO NOT SAY ”A DECREASE OF -20 ”. A DECREASE OF A NEGATIVE NUMBER IS A DOUBLE NEGATIVE, WHICH MEANS AN INCREASE! EITHER SAY ”A DECREASE OF $\$20$ ” OR ”A CHANGE OF $-\$20$ ”.

MATHS

CLARITY | CONFIDENCE | SUCCESS

Practical Example 2.3

Problem Statement: The temperature at noon was 25°C . At midnight, the temperature dropped to 18°C .

- (a) Find the change in temperature.
- (b) Interpret whether this change is an increase or a decrease, and state the actual amount.

Hint / Key Concept**Unified Subtraction**

- Always subtract in the fixed order: **New minus Original**.
- The negative sign automatically tells you the quantity went down.

Solution Steps

- (a) Calculate the change:

$$\begin{aligned}\text{Change} &= \text{New Temp} - \text{Original Temp} \\ &= 18^{\circ}\text{C} - 25^{\circ}\text{C} \\ &= -7^{\circ}\text{C}\end{aligned}$$

- (b) Interpret the result: Since the change is negative (-7°C), it represents a **decrease** of 7°C .

Final Answer: (a) -7°C , (b) A decrease of 7°C

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercises 2.3**Exercise 2.3.1****Part A (Multiple Choice)**

The score of a player changed from 50 points to 38 points. Find the change in score.

- (A) -12
- (B) 12
- (C) -88
- (D) 88

Part B (Conventional Question)

A bank account balance changed from \$800 to \$650.

- (a) Find the change in the bank account balance.
- (b) State whether this change is an increase or a decrease, and write down the actual amount.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.3.2**Part A (Multiple Choice)**

The temperature of a chemical solution was 15°C . If the change in temperature was -22°C , find the new temperature of the solution.

- (A) -37°C
- (B) -7°C
- (C) 7°C
- (D) 37°C

Part B (Conventional Question)

A stock price on Monday was \$40. On Tuesday, the change in price was +\$5. On Wednesday, the change in price was -\$8.

- (a) Find the stock price on Tuesday.
- (b) Find the stock price on Wednesday.
- (c) Calculate the total change in stock price from Monday to Wednesday.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.3.3

Reference: Inspired by HKCEE 2006 MA 1-6[cite: 2]

Part A (Multiple Choice)

The weight of Tom is 20% more than that of John[cite: 2]. It is given that John weighs 50 kg[cite: 2]. If Tom loses 12 kg, find the change in Tom's weight expressed relative to John's weight.

- (A) Tom and John now have the same weight.
- (B) Tom is now 2 kg lighter than John.
- (C) Tom is now 2 kg heavier than John.
- (D) Tom is now 10 kg lighter than John.

Part B (Conventional Question)

In January, the water level of a reservoir was 120 m. In February, the change in water level was -15 m. In March, the change in water level was $+8$ m.

- (a) Find the water level at the end of March.
- (b) Find the overall change in water level from January to March.
- (c) State whether the overall water level increased or decreased from January to March, and state the actual amount.

— Solution Space (Full Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

2.4 Percentage Change

Intuition Bridge: In Section 2.3, we learned that *change* is a signed quantity (New – Original). When we compare this signed change relative to the starting baseline as a percentage, we get percentage change. A positive percentage change means a percentage increase, while a negative percentage change means a percentage decrease. This single formula unifies both concepts seamlessly!

Skill 2.4: Calculating Percentage Change

We establish the unified mathematical formula for percentage change.

- **Definition:** Percentage change measures the change in a quantity relative to its original value, expressed as a percentage:

$$\text{Percentage Change} = \frac{\text{Change}}{\text{Original Value}} \times 100\% = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100\%$$

- **Interpretation:**

- If Percentage Change > 0 , it is a **percentage increase**.
- If Percentage Change < 0 , it is a **percentage decrease**.

- **New Value Formula:**

$$\text{New Value} = \text{Original Value} \times (1 + \text{Percentage Change})$$

TRAP WARNING: TRAP: ALWAYS PLACE THE ORIGINAL VALUE IN THE DENOMINATOR! NEVER USE THE NEW VALUE! ALSO, A PERCENTAGE CHANGE OF -20% MEANS A DECREASE OF 20% , NOT A DECREASE OF -20% !

MATHS

CLARITY | CONFIDENCE | SUCCESS

Practical Example 2.4

Problem Statement: The attendance at a concert dropped from 500 people on Friday to 400 people on Saturday.

- Find the percentage change in attendance from Friday to Saturday.
- Interpret the result in terms of percentage increase or decrease.

Hint / Key Concept**Unified Calculation**

- **Original Baseline:** Friday's attendance (500).
- **Negative Sign:** -20% directly indicates a reduction.

Solution Steps

- Calculate the percentage change using the unified formula:

$$\text{Change} = \text{New} - \text{Original} = 400 - 500 = -100$$

$$\begin{aligned}\text{Percentage Change} &= \frac{\text{Change}}{\text{Original Value}} \times 100\% \\ &= \frac{-100}{500} \times 100\% = -\mathbf{20\%}\end{aligned}$$

- Interpretation: Since the percentage change is -20% , the attendance experienced a **percentage decrease** of 20% .

Final Answer: (a) -20% , (b) A decrease of 20%

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercises 2.4**Exercise 2.4.1****Part A (Multiple Choice)**

A number changes from 80 to 60. Find the percentage change.

- (A) -25%
- (B) -20%
- (C) $+20\%$
- (D) $+25\%$

Part B (Conventional Question)

The score of a student in a test decreased from 50 to 40.

- (a) Find the change in score.
- (b) Calculate the percentage change in score.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.4.2**Part A (Multiple Choice)**

The daily wage of a worker increased from \$400 to \$480. Find the percentage change in his daily wage.

- (A) +16.7%
- (B) +20%
- (C) +25%
- (D) +80%

Part B (Conventional Question)

A bakery sold 200 cakes in May and 170 cakes in June.

- (a) Calculate the percentage change in the number of cakes sold from May to June.
- (b) In July, the number of cakes sold increased by 10% compared to June. Find the number of cakes sold in July.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 2.4.3

Reference: Inspired by HKDSE 2015-DSE-MATHS 2-9 / HKCEE Multi-stage Percentage Change[cite: 1]

Part A (Multiple Choice)

If the price of a souvenir is increased by 50% and then decreased by 20%, find the overall percentage change in the price of the souvenir.

- (A) +10%
- (B) +20%
- (C) +30%
- (D) +70%

Part B (Conventional Question)

A smartphone was originally priced at \$4000.

- (a) In January, its price was increased by 20%. Find its price in January.
- (b) In February, its January price was decreased by 25%. Find its final price in February.
- (c) Calculate the overall percentage change in price from the original price to the final February price.

— Solution Space (Full Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

3 Buying and Selling

3.1 Marked Price, Selling Price, Discount and Discount Percentage

Intuition Bridge: When you go shopping, the price printed on the price tag is the marked price. If the store has a sale, they slash that tag price by an amount called the discount. The actual cash you pay at the register is the selling price. Just like percentage change, **the marked price is always your baseline (100% denominator)** because discounts are always measured relative to the tag price!

Skill 3.1: Calculating Discount and Discount Percentage

We establish the operational formulas for marked price, selling price, discount, and discount percentage.

- **Basic Relationship:**

$$\text{Discount} = \text{Marked Price} - \text{Selling Price}$$

- **Discount Percentage Formula:**

$$\text{Discount Percentage} = \frac{\text{Discount}}{\text{Marked Price}} \times 100\%$$

- **Direct Selling Price Formula:** To find the selling price directly after a discount of $d\%$:

$$\text{Selling Price} = \text{Marked Price} \times (1 - d\%)$$

TRAP WARNING: TRAP: DISCOUNT PERCENTAGE IS ALWAYS CALCULATED RELATIVE TO THE MARKED PRICE! NEVER DIVIDE THE DISCOUNT BY THE SELLING PRICE!

MATHS

CLARITY | CONFIDENCE | SUCCESS

Practical Example 3.1

Problem Statement: A pair of sneakers has a marked price of \$500. During a seasonal sale, it is sold at a discount of 15%.

- (a) Find the discount in dollars.
- (b) Find the selling price of the sneakers.

Hint / Key Concept**Shopping Logic**

- **Marked Price:** \$500 (100% baseline).
- **Discount:** 15% off means you pay $(100 - 15)\% = 85\%$ of the marked price.

Solution Steps

- (a) Calculate the discount amount:

$$\begin{aligned}\text{Discount} &= \text{Marked Price} \times \text{Discount \%} \\ &= \$500 \times 15\% \\ &= \$500 \times \frac{15}{100} = \mathbf{\$75}\end{aligned}$$

- (b) Calculate the selling price:

$$\begin{aligned}\text{Selling Price} &= \text{Marked Price} - \text{Discount} \\ &= \$500 - \$75 = \mathbf{\$425}\end{aligned}$$

Alternative Method for (b):

$$\begin{aligned}\text{Selling Price} &= \$500 \times (1 - 15\%) \\ &= \$500 \times 0.85 = \mathbf{\$425}\end{aligned}$$

Final Answer: (a) \$75, (b) \$425

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercises 3.1**Exercise 3.1.1****Part A (Multiple Choice)**

The marked price of a shirt is \$100. If it is sold for \$80, find the discount percentage.

- (A) 10%
- (B) 20%
- (C) 25%
- (D) 80%

Part B (Conventional Question)

The marked price of a pencil case is \$60. It is sold at a discount of \$12.

- (a) Find the selling price of the pencil case.
- (b) Calculate the discount percentage.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 3.1.2**Part A (Multiple Choice)**

A jacket with a marked price of \$400 is sold at a 25% discount. Find its selling price.

- (A) \$100
- (B) \$250
- (C) \$300
- (D) \$375

Part B (Conventional Question)

The marked price of a game console is \$2400. During a festival promotion, it is offered at a discount of 18%.

- (a) Find the discount amount in dollars.
- (b) Find the selling price of the game console.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 3.1.3

Reference: Inspired by HKCEE 1998 MA 1-7 / HKDSE MA 2019-1-5[cite: 2]

Part A (Multiple Choice)

A handbag is sold at a discount of 20% on its marked price[cite: 2]. If the selling price of the handbag is \$480, find its marked price.

- (A) \$384
- (B) \$576
- (C) \$600
- (D) \$640

Part B (Conventional Question)

The marked price of a vase is \$400[cite: 2]. The vase is sold at a discount of 20% on its marked price[cite: 2].

- (a) Find the selling price of the vase[cite: 2].
- (b) If a member gets an additional cash discount of \$20 off the discounted selling price, calculate the overall discount percentage based on the original marked price.

— Solution Space (Full Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

3.2 Cost, Selling Price, Profit/Loss and Profit/Loss Percentage

Intuition Bridge: When a shopkeeper runs a business, they first spend money to buy goods. This original money spent is the cost price (or simply *cost*). When they sell the item to a customer, the money received is the selling price. If they receive more than they spent, they make a profit; if they receive less, they suffer a loss. Crucially, **the cost price is always your baseline (100% denominator)** because business gain or loss is measured relative to what was originally spent!

Skill 3.2: Profit, Loss, and Profit/Loss Percentage

We establish the core financial definitions for profit, loss, and their corresponding percentages.

- **Basic Relationships:**

When Selling Price > Cost Price : Profit = Selling Price – Cost Price

When Selling Price < Cost Price : Loss = Cost Price – Selling Price

- **Percentage Formulas:**

$$\text{Profit Percentage} = \frac{\text{Profit}}{\text{Cost Price}} \times 100\%$$

$$\text{Loss Percentage} = \frac{\text{Loss}}{\text{Cost Price}} \times 100\%$$

- **Direct Selling Price Formulas:**

$$\text{Selling Price} = \text{Cost Price} \times (1 + \text{Profit } \%)$$

$$\text{Selling Price} = \text{Cost Price} \times (1 - \text{Loss } \%)$$

TRAP WARNING: TRAP: PROFIT PERCENTAGE AND LOSS PERCENTAGE ARE ALWAYS CALCULATED RELATIVE TO THE COST PRICE! NEVER DIVIDE BY THE SELLING PRICE!

MATHS

CLARITY | CONFIDENCE | SUCCESS

Practical Example 3.2

Problem Statement: A shop owner buys a toy for \$80 and sells it to a customer for \$100.

- (a) Find the profit in dollars.
- (b) Find the profit percentage.

Hint / Key Concept**Business Baseline**

- **Cost Price:** \$80 is the 100% base.
- **Denominator:** Divide by cost price (\$80), NOT selling price (\$100).
- **Check:** \$20 gain on an \$80 investment represents a 25% gain.

Solution Steps

- (a) Calculate the profit:

$$\text{Profit} = \text{Selling Price} - \text{Cost Price}$$

$$= \$100 - \$80 = \mathbf{\$20}$$

- (b) Calculate the profit percentage:

$$\text{Profit Percentage} = \frac{\text{Profit}}{\text{Cost Price}} \times 100\%$$

$$= \frac{20}{80} \times 100\%$$

$$= \frac{1}{4} \times 100\% = \mathbf{25\%}$$

Final Answer: (a) \$20, (b) 25%

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercises 3.2**Exercise 3.2.1****Part A (Multiple Choice)**

An item bought for \$50 is sold for \$60. Find the profit percentage.

- (A) 10%
- (B) 16.7%
- (C) 20%
- (D) 83.3%

Part B (Conventional Question)

A merchant bought a sweater for \$200 and sold it for \$160.

- (a) Find the loss in dollars.
- (b) Calculate the loss percentage.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 3.2.2**Part A (Multiple Choice)**

The cost price of a textbook is \$120. If it is sold at a profit percentage of 25%, find its selling price.

- (A) \$30
- (B) \$90
- (C) \$145
- (D) \$150

Part B (Conventional Question)

A hawker bought 10 boxes of fruit at \$80 per box. He sold 8 boxes at \$100 per box, but the remaining 2 boxes were damaged and sold at a discounted price of \$30 per box.

- (a) Find the total cost price and the total selling price of all 10 boxes.
- (b) Determine whether the hawker made an overall profit or loss, and calculate the overall profit or loss percentage.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 3.2.3

Reference: Inspired by HKCEE 1990 MA 1-1 / HKCEE 2005 MA 1-6[cite: 2]

Part A (Multiple Choice)

The cost of a calculator is \$160[cite: 2]. If the calculator is sold at a profit percentage of 25%, find its selling price[cite: 2].

- (A) \$120
- (B) \$185
- (C) \$200
- (D) \$220

Part B (Conventional Question)

A person bought 10 gold coins at \$3000 each and later sold them all at \$2700 each[cite: 2].

- (a) Find the total loss made on selling all 10 gold coins[cite: 2].
- (b) Find the percentage loss[cite: 2].

— Solution Space (Full Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

3.3 Combined Scenario

Intuition Bridge: In real life, shopkeeper decisions happen in stages: first, they set a marked price higher than the cost price (this is called *markup*). Next, to attract customers, they offer a discount off that marked price to arrive at the final selling price. To find out if the store actually made money, always trace the flow step-by-step: **Cost → Marked Price → Selling Price → Profit/Loss.**

Skill 3.3: Connecting Cost, Marked Price, and Selling Price

We establish the multi-stage chain connecting all three major prices in buying and selling.

- **The Three-Price Chain:**

$$\text{Cost Price} \xrightarrow{\text{Markup } (1+m\%)} \text{Marked Price} \xrightarrow{\text{Discount } (1-d\%)} \text{Selling Price}$$

- **Stage 1 (Cost to Marked Price):**

$$\text{Marked Price} = \text{Cost Price} \times (1 + \text{Markup } \%)$$

- **Stage 2 (Marked Price to Selling Price):**

$$\text{Selling Price} = \text{Marked Price} \times (1 - \text{Discount } \%)$$

- **Stage 3 (Overall Profit / Loss Evaluation):**

$$\text{Overall Profit } \% = \frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}} \times 100\%$$

TRAP WARNING: TRAP: MARKUP PERCENTAGE IS CALCULATED RELATIVE TO COST PRICE, WHILE DISCOUNT PERCENTAGE IS CALCULATED RELATIVE TO MARKED PRICE! NEVER SUBTRACT DISCOUNT PERCENTAGE DIRECTLY FROM MARKUP PERCENTAGE (E.G., A 20% MARKUP FOLLOWED BY A 20% DISCOUNT DOES NOT MEAN 0% PROFIT)!

MATHS

CLARITY | CONFIDENCE | SUCCESS

Practical Example 3.3

Problem Statement: A handbag costs \$400. The marked price of the handbag is set 20% above its cost. During a sale, it is sold at a 25% discount on its marked price.

- Find the marked price of the handbag.
- Find the selling price of the handbag.
- Determine the overall profit or loss percentage.

Hint / Key Concept**Two Different Bases!**

- **Stage 1 Base:** \$400 (Cost).
- **Stage 2 Base:** \$480 (Marked Price).
- Even though markup was 20% and discount was 25%, the baseline grew before discount was applied!

Solution Steps

- (a) Calculate Marked Price (Markup on Cost):

$$\begin{aligned}\text{Marked Price} &= \$400 \times (1 + 20\%) \\ &= \$400 \times 1.2 = \mathbf{\$480}\end{aligned}$$

- (b) Calculate Selling Price (Discount on Marked Price):

$$\begin{aligned}\text{Selling Price} &= \$480 \times (1 - 25\%) \\ &= \$480 \times 0.75 = \mathbf{\$360}\end{aligned}$$

- (c) Calculate Overall Profit/Loss Percentage:

$$\text{Profit/Loss} = \$360 - \$400 = -\$40 \quad (\text{Loss})$$

$$\text{Loss \%} = \frac{\$40}{\$400} \times 100\% = \mathbf{10\%}$$

Final Answer: (a) \$480, (b) \$360, (c) A loss of 10%

CLARITY | CONFIDENCE | SUCCESS

Exercises 3.3**Exercise 3.3.1****Part A (Multiple Choice)**

An article with a cost price of \$100 is marked 30% above cost. It is then sold at a 10% discount on its marked price. Find its selling price.

- (A) \$117
- (B) \$120
- (C) \$127
- (D) \$130

Part B (Conventional Question)

An item costs \$200. The marked price is set 50% higher than the cost price. It is later sold at a 20% discount on the marked price.

- (a) Find the marked price of the item.
- (b) Find the selling price of the item.
- (c) Find the overall profit or loss percentage.

— Solution Space (Half Page) —

MATHS

CLARITY | CONFIDENCE | SUCCESS

Exercise 3.3.2**Part A (Multiple Choice)**

The marked price of a toy is 40% higher than its cost price. If the toy is sold at a 20% discount on its marked price, what is the overall profit percentage based on cost?

- (A) 8%
- (B) 12%
- (C) 20%
- (D) 32%

Part B (Conventional Question)

A merchant bought a chair for \$360. He set the marked price such that if he offered a 20% discount on the marked price, he would still make a profit percentage of 30% on cost.

- (a) Find the selling price of the chair.
- (b) Calculate the marked price of the chair.

— Solution Space (Half Page) —

MATHS

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Exercise 3.3.3

Reference: Inspired by HKCEE 1994 MA 1-6 / HKCEE 2003 MA 1-5 / HKDSE PP-1-4[cite: 2]

Part A (Multiple Choice)

A merchant bought an article for $\$x$. He put it in his shop for sale at a marked price 70% higher than its cost[cite: 2]. The article was then sold to a customer at a discount of 5%[cite: 2]. Find the overall percentage gain for the merchant[cite: 2].

- (A) 61.5%
- (B) 65.0%
- (C) 66.5%
- (D) 73.5%

Part B (Conventional Question)

The marked price of a birthday cake is $\$360$ [cite: 2]. The birthday cake is sold at a discount of 45% on its marked price[cite: 2].

- (a) Find the selling price of the birthday cake[cite: 2].
- (b) If the marked price of the birthday cake is 80% above its cost, determine whether there will be a profit or a loss after selling the birthday cake[cite: 2]. Explain your answer with detailed calculations[cite: 2].

— Solution Space (Full Page) —

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4 Interest Rate

4.1 Simple Interest

Intuition Bridge: When you store money in a bank (or borrow money from a lender), a fee is paid for using that money, called interest. Under *simple interest*, the interest fee earned each year is calculated **only on the original starting money** (the principal). This means the dollar amount of interest added in Year 1, Year 2, and Year 3 is **always identical!**

Skill 4.1: Calculating Simple Interest and Total Amount

We establish the core financial definitions and formulas for simple interest.

- **Basic Definitions:**

- Principal (P): The original sum of money deposited or borrowed.
- Interest Rate per annum ($r\%$ p.a.): The percentage of principal charged or earned per year.
- Time period (T): The duration in **years**.

- **Simple Interest Formula:**

$$I = P \times r\% \times T = \frac{P \cdot r \cdot T}{100}$$

- **Total Amount Formula:** The total accumulated sum (A) after T years is:

$$A = P + I = P(1 + r\% \cdot T)$$

TRAP WARNING: TRAP: TIME T MUST ALWAYS BE EXPRESSED IN YEARS! IF TIME IS GIVEN IN MONTHS, YOU MUST CONVERT IT TO YEARS BY DIVIDING BY 12 (E.G., 6 MONTHS = $\frac{6}{12} = 0.5$ YEARS)!

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Practical Example 4.1

Problem Statement: Mr. Wong deposits \$10 000 into a bank at a simple interest rate of 5% per annum for 3 years.

- (a) Find the simple interest earned.
 (b) Find the total amount he receives after 3 years.

Hint / Key Concept**Constant Annual Interest**

- **1 Year Interest:** $\$10\,000 \times 5\% = \500 .
- **3 Years Interest:** $\$500 \times 3 = \$1\,500$.
- Under simple interest, the principal baseline never changes.

Solution Steps

- (a) Calculate simple interest I :

$$P = \$10\,000, \quad r\% = 5\%, \quad T = 3$$

$$\begin{aligned} I &= P \times r\% \times T \\ &= \$10\,000 \times \frac{5}{100} \times 3 = \mathbf{\$1\,500} \end{aligned}$$

- (b) Calculate total amount A :

$$\begin{aligned} A &= P + I \\ &= \$10\,000 + \$1\,500 = \mathbf{\$11\,500} \end{aligned}$$

Final Answer: (a) \$1 500, (b) \$11 500

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Exercises 4.1**Exercise 4.1.1****Part A (Multiple Choice)**

Find the simple interest earned on a principal of \$2 000 at an interest rate of 4% per annum for 2 years.

- (A) \$80
- (B) \$160
- (C) \$2 160
- (D) \$2 320

Part B (Conventional Question)

A deposit of \$5 000 is placed in a bank at a simple interest rate of 3% per annum for 4 years.

- (a) Calculate the simple interest earned.
- (b) Calculate the total amount accumulated at the end of 4 years.

— Solution Space (Half Page) —

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Exercise 4.1.2**Part A (Multiple Choice)**

Find the simple interest on \$6 000 at 6% per annum for 6 months.

- (A) \$180
- (B) \$360
- (C) \$1 800
- (D) \$3 600

Part B (Conventional Question)

A man borrows \$8 000 from a finance company at a simple interest rate of 5% per annum. He promises to repay the loan together with interest after 18 months.

- (a) Express 18 months in terms of years.
- (b) Find the total simple interest he needs to pay.
- (c) Calculate the total amount required to fully settle the loan.

— Solution Space (Half Page) —

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Exercise 4.1.3

Reference: Inspired by HKCEE 1977 CE MATHS 2-4 / HKCEE 1980 CE MATHS 2-33[?]

Part A (Multiple Choice)

If \$ P amounts to \$ Q in n years at simple interest, the rate per annum is[?]

- (A) $\frac{100n(Q-P)}{P}\%$ [?]
- (B) $\frac{100(Q-P)}{n}\%$ [?]
- (C) $\frac{100(Q-P)}{nP}\%$ [?]
- (D) $\frac{100(Q-P)}{nQ}\%$ [?]

Part B (Conventional Question)

Mr. Wong puts \$10 000 into Bank X and he also puts \$10 000 into Bank Y[?]. The simple interest per year from Bank X is \$50 more than that from Bank Y[?]. It is given that the interest rate of Bank Y is 5% per annum[?].

- (a) Find the simple interest received from Bank Y in one year[?].
- (b) Find the simple interest received from Bank X in one year[?].
- (c) Determine the simple interest rate per annum offered by Bank X[?].

— Solution Space (Full Page) —

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4.2 Compound Interest

Intuition Bridge: Unlike simple interest where the principal stays fixed forever, compound interest is “**interest on interest**”. At the end of each compounding period, the interest earned is added back to the principal to form a new, larger principal for the next period. It’s like a snowball rolling down a hill: **the bigger it gets, the faster it grows!**

Skill 4.2: Compound Amount and Compound Interest

We establish the operational formulas for compound interest and compound amount.

- **Basic Concepts:**

- Principal (P): The original amount deposited or borrowed.
- Compounding Period: The time interval after which interest is added back to the principal (e.g., yearly, half-yearly, quarterly, monthly).
- Periodic Interest Rate: The interest rate per compounding period.
- Total Number of Compounding Periods (N): Number of compounding cycles over the entire duration.

- **Compound Amount Formula:**

$$\text{Compound Amount } A = P \left(1 + \frac{r\%}{k} \right)^{n \cdot k}$$

where $r\%$ is the annual rate, n is the number of years, and k is the number of compounding periods per year.

- **Compound Interest Formula:**

$$\text{Compound Interest } I = A - P = P \left[\left(1 + \frac{r\%}{k} \right)^{n \cdot k} - 1 \right]$$

TRAP WARNING: TRAP: DO NOT CONFUSE THE ANNUAL INTEREST RATE WITH THE PERIODIC INTEREST RATE! IF INTEREST IS COMPOUNDED MONTHLY, DIVIDE THE ANNUAL RATE BY 12 AND MULTIPLY THE NUMBER OF YEARS BY 12!

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Practical Example 4.2

Problem Statement: \$10 000 is deposited into a bank for 2 years at an interest rate of 10% per annum, compounded yearly.

- (a) Find the total compound amount at the end of 2 years.
 (b) Find the compound interest earned.

Hint / Key Concept**Year-by-Year Growth**

- **Year 1 End:** $\$10\,000 \times 1.1 = \$11\,000$.
- **Year 2 Start:** New principal is \$11 000.
- **Year 2 End:** $\$11\,000 \times 1.1 = \$12\,100$.
- Notice that Year 2 earns \$1 100 interest, compared to \$1 000 in Year 1!

Solution Steps

- (a) Calculate compound amount A :

$$P = \$10\,000, \quad r\% = 10\%, \quad n = 2$$

$$\begin{aligned} A &= P(1 + r\%)^n \\ &= \$10\,000 \times (1 + 10\%)^2 \\ &= \$10\,000 \times (1.1)^2 \\ &= \$10\,000 \times 1.21 = \mathbf{\$12\,100} \end{aligned}$$

- (b) Calculate compound interest I :

$$\begin{aligned} I &= A - P \\ &= \$12\,100 - \$10\,000 = \mathbf{\$2\,100} \end{aligned}$$

Final Answer: (a) \$12 100, (b) \$2 100

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Exercises 4.2**Exercise 4.2.1****Part A (Multiple Choice)**

Find the total amount when \$10 000 is deposited at 4% per annum for 2 years, compounded yearly.

- (A) \$10 800
- (B) \$10 816
- (C) \$11 600
- (D) \$11 664

Part B (Conventional Question)

A sum of \$5 000 is deposited in a bank at an interest rate of 2% per annum, compounded yearly for 3 years.

- (a) Find the total amount accumulated at the end of 3 years (correct to the nearest dollar).
- (b) Calculate the total compound interest earned.

— Solution Space (Half Page) —

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Exercise 4.2.2**Part A (Multiple Choice)**

A sum of \$10 000 is deposited at an interest rate of 12% per annum for 1 year, compounded monthly. Find the interest rate per month and the total number of compounding periods.

- (A) Monthly rate = 1%, Total periods = 12
- (B) Monthly rate = 1%, Total periods = 1
- (C) Monthly rate = 12%, Total periods = 12
- (D) Monthly rate = 6%, Total periods = 2

Part B (Conventional Question)

\$20 000 is deposited in a bank for 2 years at an interest rate of 8% per annum, compounded half-yearly.

- (a) Find the interest rate per half-year and the total number of half-years.
- (b) Calculate the compound amount correct to the nearest dollar.
- (c) Find the compound interest earned correct to the nearest dollar.

— Solution Space (Half Page) —

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Exercise 4.2.3

Reference: Inspired by HKCEE 1992-CE-MATHS 2-12 / HKDSE 2015-DSE-MATHS 2-10[?]

Part A (Multiple Choice)

A sum of \$50 000 is deposited at an interest rate of 6% per annum for 3 years, compounded quarterly[?]. Find the amount correct to the nearest dollar[?].

- (A) \$59 000[?]
- (B) \$59 551[?]
- (C) \$59 755[?]
- (D) \$59 781[?]

Part B (Conventional Question)

A sum of \$10 000 is deposited at an interest rate of 4% per annum, compounded yearly[?].

- (a) Find the total amount accumulated at the end of the first year.
- (b) Find the total amount accumulated at the end of the second year.
- (c) Calculate the interest earned **in the second year alone**[?].

— Solution Space (Full Page) —

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4.3 Tax Computation with Stages

Intuition Bridge: When citizens earn money, they pay salaries tax to the government. Instead of charging a single flat rate on all your earnings, governments often use a **progressive tax system**. Think of your taxable income as water poured into a series of buckets (tax bands): the first bucket is charged at a low percentage, the next bucket at a higher percentage, and so on. You only pay higher rates on the money that spills into higher buckets!

Skill 4.3: Salaries Tax and Progressive Tax Bands

We establish the step-by-step procedure for computing progressive salaries tax.

- **Step 1: Calculate Net Chargeable Income:**

$$\text{Net Chargeable Income} = \text{Total Annual Income} - \text{Total Allowances}$$

- **Step 2: Fill the Tax Bands (Progressive Rates):**

- Tax is calculated band-by-band up to the width of each bracket.
- Tax for Band $i = \text{Income in Band } i \times \text{Tax Rate}_i$

- **Step 3: Sum the Tax Amounts:**

$$\text{Total Salaries Tax Payable} = \text{Tax}_{\text{Band 1}} + \text{Tax}_{\text{Band 2}} + \cdots + \text{Tax}_{\text{Remaining}}$$

TRAP WARNING: TRAP: TAX RATES ARE ONLY APPLIED TO THE AMOUNT OF INCOME THAT FALLS WITHIN THAT SPECIFIC BAND! NEVER MULTIPLY YOUR ENTIRE INCOME BY THE HIGHEST TAX RATE BAND! ALWAYS SUBTRACT ALLOWANCES FIRST BEFORE CALCULATING TAX!

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Practical Example 4.3

Problem Statement: Mr. Lee has an annual income of \$220 000. The basic allowance is \$120 000. The progressive tax rates are as follows:

Net Chargeable Income	Tax Rate
First \$40 000	2%
Next \$40 000	7%
Remainder	12%

Find the total salaries tax payable by Mr. Lee.

Hint / Key Concept**Bucket Analogy**

- **Total Water:** \$100 000 after allowance deduction.
- **Bucket 1:** Holds \$40 000 (Tax = \$800).
- **Bucket 2:** Holds \$40 000 (Tax = \$2 800).
- **Bucket 3:** Holds remaining \$20 000 (Tax = \$2 400).

Solution Steps

1. Calculate Net Chargeable Income:

$$\text{Income} = \$220\,000 - \$120\,000 = \mathbf{\$100\,000}$$

2. Split \$100 000 into tax bands:

- First \$40 000: $\$40\,000 \times 2\% = \800
- Next \$40 000: $\$40\,000 \times 7\% = \$2\,800$
- Remaining ($\$100\,000 - \$80\,000 = \$20\,000$):

$$\$20\,000 \times 12\% = \$2\,400$$

3. Calculate total tax payable:

$$\text{Total Tax} = \$800 + \$2\,800 + \$2\,400 = \mathbf{\$6\,000}$$

Final Answer: \$6 000

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Exercises 4.3**Exercise 4.3.1****Part A (Multiple Choice)**

The net chargeable income of a taxpayer is \$60 000. The tax schedule is: First \$30 000 at 2%, Remaining income at 8%. Find the tax payable.

- (A) \$1 200
- (B) \$2 400
- (C) \$3 000
- (D) \$4 800

Part B (Conventional Question)

The net chargeable income of Miss Cheung is \$90 000. The tax rates are structured as follows:

Net Chargeable Income	Tax Rate
First \$50 000	2%
Remainder	6%

- (a) Calculate the tax charged on the first \$50 000.
- (b) Calculate the tax charged on the remaining income.
- (c) Find the total salaries tax payable by Miss Cheung.

— Solution Space (Half Page) —

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Exercise 4.3.2**Part A (Multiple Choice)**

Mr. Ho's annual income is \$200 000 and his total allowance is \$132 000. If the first \$50 000 of net chargeable income is taxed at 2% and the remainder at 6%, find his salaries tax payable.

- (A) \$1 000
- (B) \$1 080
- (C) \$2 080
- (D) \$4 080

Part B (Conventional Question)

In a tax year, Mr. Kwok earns an annual salary of \$350 000. He is entitled to a basic allowance of \$132 000 and a child allowance of \$120 000. The tax rates are:

Net Chargeable Income	Tax Rate
First \$50 000	2%
Next \$50 000	6%
Remainder	10%

- (a) Find Mr. Kwok's net chargeable income.
- (b) Compute the total salaries tax payable by Mr. Kwok.

— Solution Space (Half Page) —

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Exercise 4.3.3

Reference: Inspired by HKCEE 1993 CE MATHS 1-4 / HKDSE 2018 DSE MATHS 1-8

Part A (Multiple Choice)

Mrs. Li has an annual income of \$480 000. She is granted a total allowance of \$220 000. The progressive tax rate schedule is:

Net Chargeable Income	Tax Rate
First \$50 000	2%
Next \$50 000	6%
Next \$50 000	10%
Remainder	14%

Find the salaries tax payable by Mrs. Li.

- (A) \$19 400
- (B) \$23 400
- (C) \$24 800
- (D) \$36 400

Part B (Conventional Question)

Mr. Wong's monthly income is \$30 000. He is granted a basic allowance of \$132 000 for the year.

Net Chargeable Income	Tax Rate
First \$50 000	2%
Next \$50 000	7%
Next \$50 000	12%
Remainder	17%

- (a) Calculate Mr. Wong's annual income.
- (b) Find his net chargeable income for the year.
- (c) Calculate the salaries tax payable by Mr. Wong.
- (d) Express the tax payable as a percentage of his total annual income, correct to 1 decimal place.

— Solution Space (Full Page) —

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5 Growth, Decay and Successive Changes

5.1 Growth and Decay

Intuition Bridge: Whether it is a country's population expanding, bacteria multiplying in a lab, or a brand new car losing value over time (*depreciation*), real-world quantities often change by a fixed percentage rate over consecutive time periods. Notice how this mechanism works exactly like compound interest: **each period's change is calculated on the NEW value from the previous period**, compounding the growth or decay!

Skill 5.1: Calculating Constant Growth and Decay (Depreciation)

We establish the master formulas for uniform growth and decay across n time periods.

- **Growth Formula:** If a quantity P grows at a constant rate of $r\%$ per period for n periods, the final amount A is:

$$A = P(1 + r\%)^n$$

- **Decay / Depreciation Formula:** If a quantity P decays or depreciates at a constant rate of $r\%$ per period for n periods, the final amount A is:

$$A = P(1 - r\%)^n$$

- **Total Change Amount:**

$$\text{Total Growth / Depreciation} = |A - P|$$

TRAP WARNING: TRAP: FOR DECAY OR DEPRECIATION, ALWAYS USE $(1 - r\%)$, NEVER ADD! WHEN WORKING BACKWARDS TO FIND THE INITIAL VALUE P , DO NOT CALCULATE $A \times (1 + r\%)^n$; INSTEAD, SET UP THE EQUATION $P(1 \pm r\%)^n = A$ AND SOLVE FOR P !

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Practical Example 5.1

Problem Statement: A car was bought for \$200 000. Its value depreciates at a constant rate of 15% per year.

- (a) Find the value of the car after 3 years.
 (b) Find the total amount of value lost over the 3 years.

Hint / Key Concept**Compounding Effect**

- **Year 1 Value:** $\$200\,000 \times 0.85 = \$170\,000$.
- **Year 2 Value:** $\$170\,000 \times 0.85 = \$144\,500$.
- **Year 3 Value:** $\$144\,500 \times 0.85 = \$122\,825$.
- The amount lost decreases each year because 15% is applied to a smaller base!

Solution Steps

- (a) Calculate depreciated value A :

$$P = \$200\,000, \quad r\% = 15\%, \quad n = 3$$

$$A = P(1 - r\%)^n$$

$$= \$200\,000 \times (1 - 15\%)^3$$

$$= \$200\,000 \times (0.85)^3$$

$$= \$200\,000 \times 0.614125 = \textbf{\$122 825}$$

- (b) Calculate total value lost:

$$\text{Total Loss} = P - A$$

$$= \$200\,000 - \$122\,825 = \textbf{\$77 175}$$

Final Answer: (a) \$122 825, (b) \$77 175

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Exercises 5.1**Exercise 5.1.1****Part A (Multiple Choice)**

The population of a city is 100 000. It increases at a rate of 5% per year. Find the population after 2 years.

- (A) 105 000
- (B) 110 000
- (C) 110 250
- (D) 115 762

Part B (Conventional Question)

A computer system valued at \$16 000 depreciates at a rate of 20% per year.

- (a) Find its value after 1 year.
- (b) Find its value after 3 years.
- (c) Calculate the total depreciation in value after 3 years.

— Solution Space (Half Page) —

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Exercise 5.1.2**Part A (Multiple Choice)**

The value of a machine depreciates by 10% each year. If its current value is \$81 000, what was its value 2 years ago?

- (A) \$65 610
- (B) \$97 200
- (C) \$98 010
- (D) \$100 000

Part B (Conventional Question)

The number of members in a fitness club grows at a constant rate of 10% per year. At the end of 2024, the club had 2 420 members.

- (a) Find the number of members at the end of 2022 (2 years ago).
- (b) Estimate the number of members at the end of 2026 (2 years later).

— Solution Space (Half Page) —

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Exercise 5.1.3

Reference: Inspired by HKCEE 1996 CE MATHS 2-8 / HKDSE 2012 DSE MATHS 2-10

Part A (Multiple Choice)

The price of a house increases by 10% in the first year and then decreases by 10% in the second year. If the original price was $\$P$, what is the final price after 2 years?

- (A) $\$0.99P$
- (B) $\$1.00P$
- (C) $\$1.01P$
- (D) $\$1.21P$

Part B (Conventional Question)

The value of a mobile phone depreciates at a constant rate of $r\%$ per year. The initial value of the phone was $\$8\,000$. After 2 years, its value becomes $\$5\,120$.

- (a) Express the value after 2 years in terms of r .
- (b) Find the annual depreciation rate $r\%$.
- (c) Using the value of r found in (b), determine the value of the phone after 4 years.

— Solution Space (Full Page) —

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5.2 Successive Percentage Changes

Intuition Bridge: When a quantity undergoes two or more percentage changes one after another, it is tempting to simply add or subtract the percentages together. However, **percentage changes are multiplicative, not additive!** Each consecutive change is calculated using the new value after the previous change as its fresh starting base.

Skill 5.2: Calculating Successive Percentage Changes

We establish the formula for compounding multiple consecutive percentage changes.

- **Final Value Formula:** If an initial amount P undergoes successive percentage changes of $x_1\%$, $x_2\%$, $\dots$, $x_n\%$:

$$\text{Final Value } A = P \times (1 + x_1\%) \times (1 + x_2\%) \times \dots \times (1 + x_n\%)$$

Note: If a change is a decrease, $x_i\%$ is negative, so use $(1 - |x_i|\%)$.

- **Overall Percentage Change Formula:**

$$\text{Overall Percentage Change} = [(1 + x_1\%)(1 + x_2\%) \dots (1 + x_n\%) - 1] \times 100\%$$

TRAP WARNING: TRAP: NEVER ADD PERCENTAGE CHANGES DIRECTLY! AN INCREASE OF 20% FOLLOWED BY A DECREASE OF 20% DOES NOT CANCEL OUT TO 0%! INSTEAD, $(1 + 20\%)(1 - 20\%) = 1.2 \times 0.8 = 0.96$, WHICH IS A 4% NET DECREASE!

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Practical Example 5.2

Problem Statement: The price of a watch is \$1 200. In May, its price increases by 20%. In June, its new price decreases by 10%.

- (a) Find the final price of the watch after June.
 (b) Find the overall percentage change in the price of the watch.

Hint / Key Concept**Step-by-Step View**

- **May Price:** $\$1\,200 \times 1.2 = \$1\,440$.
- **June Price:** $\$1\,440 \times 0.9 = \$1\,296$.
- **Net Factor:** $1.2 \times 0.9 = 1.08$, which represents an overall 8% increase!

Solution Steps

- (a) Calculate final price A :

$$P = \$1\,200, \quad x_1\% = +20\%, \quad x_2\% = -10\%$$

$$A = P \times (1 + 20\%) \times (1 - 10\%)$$

$$= \$1\,200 \times 1.2 \times 0.9 = \mathbf{\$1\,296}$$

- (b) Calculate overall percentage change:

$$\text{Overall Change} = \frac{A - P}{P} \times 100\%$$

$$= \frac{\$1\,296 - \$1\,200}{\$1\,200} \times 100\%$$

$$= \frac{\$96}{\$1\,200} \times 100\% = \mathbf{+8\%}$$

Final Answer: (a) \$1 296, (b) +8% (or an 8% increase)

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Exercises 5.2**Exercise 5.2.1****Part A (Multiple Choice)**

A number is first increased by 10% and then further increased by 30%. What is the overall percentage increase?

- (A) 40%
- (B) 41%
- (C) 43%
- (D) 44%

Part B (Conventional Question)

The price of a toy was originally \$200. The price was increased by 15% in November and then reduced by 20% during a December promotion.

- (a) Find the price of the toy after November.
- (b) Find the price of the toy after the December promotion.
- (c) Calculate the overall percentage change in the price of the toy.

— Solution Space (Half Page) —

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Exercise 5.2.2**Part A (Multiple Choice)**

If the length of a rectangle is increased by 20% while its width is decreased by 10%, find the percentage change in its area.

- (A) Increased by 8%
- (B) Increased by 10%
- (C) Decreased by 8%
- (D) No change

Part B (Conventional Question)

The length of a square of side x cm is increased by 10% to form a rectangle, and its width is decreased by 10%.

- (a) Express the area of the original square in terms of x .
- (b) Express the area of the new rectangle in terms of x .
- (c) Find the percentage change in the area from the square to the rectangle.

— Solution Space (Half Page) —

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Exercise 5.2.3

Reference: Inspired by HKCEE 1995 CE MATHS 2-8 / HKDSE 2017 DSE MATHS 2-11

Part A (Multiple Choice)

The production cost of an item increased by 25% in 2023 and then decreased by 20% in 2024. How does the production cost at the end of 2024 compare with the original cost at the beginning of 2023?

- (A) Decreased by 5%
- (B) Unchanged (0% change)
- (C) Increased by 5%
- (D) Increased by 10%

Part B (Conventional Question)

A company's annual profit in 2022 was \$500 000. In 2023, the profit increased by $r\%$. In 2024, the profit decreased by $r\%$. It is given that the profit in 2024 was \$480 000.

- (a) Express the profit in 2024 in terms of r .
- (b) Solve for r .
- (c) Calculate the profit made in 2023.

— Solution Space (Full Page) —

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5.3 Repayment Problems and Iterative Changes

Intuition Bridge: Real-world financial situations like bank loans, mortgages, or regular savings plans involve two competing forces at work simultaneously: **percentage growth/decay** (interest accruing) and **fixed addition/subtraction** (regular repayments or deposits). Because interest is recalculated at each step on the outstanding balance, **you must track the balance period-by-period rather than subtracting all repayments at once!**

Skill 5.3: Step-by-Step Balance Tracking for Repayment

We establish the iterative method for calculating balances subject to periodic interest and constant repayments or withdrawals.

- **Basic Iterative Pattern (Loan Repayment / Withdrawal):** Let P_0 be the initial principal, $r\%$ be the interest rate per period, and R be the regular repayment made at the end of each period.

– **End of Period 1:**

$$P_1 = P_0(1 + r\%) - R$$

– **End of Period 2:**

$$P_2 = P_1(1 + r\%) - R = [P_0(1 + r\%) - R](1 + r\%) - R$$

– **End of Period n :**

$$P_n = P_{n-1}(1 + r\%) - R$$

- **Key Principle:** Interest for any period is charged only on the outstanding balance at the start of that period!

TRAP WARNING: TRAP: NEVER DEDUCT ALL REPAYMENTS FROM THE INITIAL PRINCIPAL BEFORE CALCULATING COMPOUND INTEREST! DO NOT COMPUTE $P_0(1 + r\%)^n - n \times R$ DIRECTLY UNLESS YOU FULLY EXPAND AND APPLY INTEREST TO EACH INTERMEDIATE BALANCE!

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Practical Example 5.3

Problem Statement: Mr. Chan borrows \$100 000 from a bank at an interest rate of 10% per annum, compounded yearly. At the end of each year, he repays a fixed amount of \$40 000.

- (a) Find the amount Mr. Chan owes the bank at the end of Year 1 (after the first repayment).
 (b) Find the amount Mr. Chan owes the bank at the end of Year 2 (after the second repayment).

Hint / Key Concept**Sequential Tracking**

- **Year 1:** Interest is \$10 000 because the starting principal is \$100 000.
- **Year 2:** Interest drops to \$7 000 because the principal was reduced to \$70 000!
- Notice that interest shrinks after each repayment.

Solution Steps

- (a) Balance at end of Year 1 (P_1):

$$\text{Interest for Year 1} = \$100\,000 \times 10\% = \$10\,000$$

$$\text{Balance before repayment} = \$100\,000 + \$10\,000 = \$110\,000$$

$$P_1 = \$110\,000 - \$40\,000 = \mathbf{\$70\,000}$$

- (b) Balance at end of Year 2 (P_2):

$$\text{Interest for Year 2} = \$70\,000 \times 10\% = \$7\,000$$

$$\text{Balance before repayment} = \$70\,000 + \$7\,000 = \$77\,000$$

$$P_2 = \$77\,000 - \$40\,000 = \mathbf{\$37\,000}$$

Final Answer: (a) \$70 000, (b) \$37 000

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Exercises 5.3**Exercise 5.3.1****Part A (Multiple Choice)**

A loan of \$10 000 is subject to an interest rate of 5% per annum, compounded yearly. If a repayment of \$3 000 is made at the end of each year, find the remaining balance at the end of 2 years.

- (A) \$4 000
- (B) \$4 500
- (C) \$4 875
- (D) \$5 000

Part B (Conventional Question)

A man borrows \$50 000 from a loan company at an interest rate of 8% per annum, compounded yearly. He agrees to repay \$20 000 at the end of each year.

- (a) Calculate the outstanding balance at the end of the first year (after the first repayment).
- (b) Calculate the outstanding balance at the end of the second year (after the second repayment).
- (c) Find the total interest charged by the loan company over these 2 years.

— Solution Space (Half Page) —

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Exercise 5.3.2**Part A (Multiple Choice)**

A savings account starts with \$200 000 and earns interest at a rate of 4% per annum, compounded yearly. At the end of each year, \$50 000 is withdrawn from the account. Find the balance in the account at the end of 2 years.

- (A) \$100 000
- (B) \$108 000
- (C) \$114 320
- (D) \$120 000

Part B (Conventional Question)

Ms. Wong takes a personal loan of \$21 000 at an interest rate of 10% per annum, compounded yearly. She repays an equal amount of \$ R at the end of each year for 2 years to fully clear the debt at the end of the second year.

- (a) Express the outstanding balance at the end of the first year in terms of R .
- (b) Express the outstanding balance at the end of the second year in terms of R .
- (c) Hence, find the value of R .

— Solution Space (Half Page) —

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Exercise 5.3.3

Reference: Inspired by HKCEE 1998 CE MATHS 2-11 / HKDSE 2019 DSE MATHS 2-12

Part A (Multiple Choice)

Peter borrows \$100 000 at an interest rate of 6% per annum, compounded yearly. He repays \$50 000 at the end of the first year. How much does he need to pay at the end of the second year to fully clear the remaining loan?

- (A) \$50 000
- (B) \$56 000
- (C) \$59 360
- (D) \$62 000

Part B (Conventional Question)

A fish farm initially contains 10 000 fish. At the end of each year, the fish population increases by 20% due to natural reproduction, and then 3 000 fish are harvested (harvested after the annual growth).

- (a) Find the number of fish in the farm at the end of Year 1 (after harvest).
- (b) Find the number of fish in the farm at the end of Year 2 (after harvest).
- (c) Find the number of fish in the farm at the end of Year 3 (after harvest).
- (d) Calculate the overall percentage change in the number of fish from the initial population to the end of Year 3.

— Solution Space (Full Page) —

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